SNAHASHIS SARKER ARNOB

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SUMMARY

Computer Science and Engineering graduate with hands-on research and industry experience in full-stack web development and artificial intelligence. Proficient in MERN-stack architectures and deep learning, with research spanning LLM memory architectures, hyperspectral computer vision and NLP-driven discourse analysis, including a co-authored manuscript under peer review. Interested in enterprise computer networking and preparing for the CCNA certification. Self-motivated fast learner with strong analytical and coding skills proven in Agile teams.

EDUCATION

North South University 2022–2026
Bachelor of Science, Computer Science and Engineering Dhaka, Bangladesh
GPA 3.40/4.00

WORK EXPERIENCE

Junior Software Engineer Sep 2025 – May 2026
Cloversoft Dhaka, Bangladesh

- Built features for a MERN-stack service management platform from client requirements set by the project lead.
- Delivered tasks in weekly Agile Scrum sprints, including bug fixes and changes from client feedback.

Junior Web Developer (Intern) Jan 2025 – Jun 2025
Qualifie Dhaka, Bangladesh

- Managed and optimized JSON data structures to drive dynamic content delivery.
- Validated schema consistency in JSON configuration files for reliable frontend rendering.

PROJECTS AND PUBLICATIONS

- **Boier Biddan** (Project)
  Full-stack Next.js app that generates illustrated Bengali children's stories from user prompts and chosen art styles, using Google Gemini for text and Black Forest Labs' FLUX (via Hugging Face) for images, with user accounts, personal dashboards, a public story feed, and ShurjoPay sandbox payments; deployed on Vercel.
- **ServeEasy** (Project)
  Solo-built full-stack Next.js on-demand home services marketplace where users discover and hire professionals for cleaning, plumbing, cooking and more, with Descope authentication.
- **Contextual Hierarchical Rolling Orchestration of Narrative Intelligence with Continuous Layered Evolution (CHRONICLE)** (Project)
  Five-tier "living summary" memory architecture for LLMs that merges new facts via contradiction-aware semantic absorption with personalized confidence decay, retrieves context top-down from persona to turn level, and uses LLM-judge validation plus provenance-weighted guards against stale and adversarially injected memories.
- **Nitrate Prediction for Southern Ocean Seagliders** (Project)
  Regression pipeline on ∼860k SOGOS Seaglider samples (median imputation, IQR outlier removal, PCA), comparing Ridge, Decision Tree, stacking, bagged Extra Trees and XGBoost tuned via halving random search; bagged Extra Trees reached $R^2 = 0.997$ vs. a 0.859 Ridge baseline.
- **Hyperspectral Ink Classification for Historical Documents** (Project)
  Hyperspectral ML pipeline for 15-class historical ink classification, with a predictive band-budget meta-model that recommends the minimum spectral bands per ink class for a target F1, enabling low-cost heritage analysis.
- **The Builder-Worker Paradox: A Mixed-Methods Analysis of Affective Dissonance in Agentic AI Discourse on Social Media** (Co-author, equal contribution; under review at *Online Social Networks and Media*, Elsevier)
  Mixed-methods NLP study analyzing 19,250 Reddit posts with BERTopic and GoEmotions, showing labor-related AI discourse carries 2× the anxiety of builder discourse ($p < 0.001$), robust to keyword masking (94.7% retention).

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, SQL, C/C++
- **Machine Learning and NLP:** scikit-learn, Keras, TensorFlow, PyTorch, NumPy, pandas
- **Data Visualization:** Matplotlib, Seaborn, Power BI
- **Networking:** Cisco Packet Tracer, Cisco IOS CLI, static routing, OSPF, RIP, DHCP, DNS, NAT, Wireshark
- **Domain Specific:** LLM memory systems & RAG, NLP & topic modeling, hyperspectral imaging, generative AI